

Critical moments in long-term condition management: A longitudinal qualitative social network study

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Abstract

Objectives: Recent literature has demonstrated the nature and importance of social networks. This study aimed to understand how support needs of people with long-term conditions change over time and how this influences their ability to self-manage.

Methods: A longitudinal qualitative design was used to explore changes in individual's social networks over a period of year. Thirty participants were recruited and completed initial in-depth face-to-face interviews, telephone follow-ups, and final face-to-face interviews.

Results: Findings illuminated that changes in health and changes in social networks can influence each other. The social networks implicated in the management of long-term conditions change over time at critical moments and can range between positive and negative reciprocal influences with self-management. Network changes, such as the breaking or reconnecting of ties influenced the context of health management and the degree of engagement with self-management activities.

Discussion: By examining the specific processes of support and resource provision, this study highlights how overtime the variety of relationships and support available implicated in long-term condition management changes. This has implications for the provision of formal support services and the need for a wider range of support to be identified that recognises change in support requirements as self-management strategies evolve.

Keywords

Social networks, longitudinal, qualitative, long-term condition management, UK

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Introduction

A key aim of global health policies is to support people to self-manage. Self-management can be defined as the activity that an individual undertakes for themselves or their family to remain healthy and care

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for acute or long-term conditions.¹ Small changes in everyday routines can have important influence on illness trajectories or individual response to management.² Implicit within the support available for long-term condition management is that either less support, or the same support, is needed over time. Whilst there is a plethora of research that has explored individual self-management at a single point in time,^{3–5} understanding how self-management changes over time has remained underexplored.

The processes of long-term condition management (such as managing medication regimes or lifestyle modifications) involving ongoing work may change over time. The context in which condition management occurs is in a constant state of flux which places increased pressure on the resources an individual has available to self-manage.² The temporal aspects of living with a long-term condition have been found to influence biographical reflections associated with illness trajectories.^{6,7} Less attention has been given to understanding the impact of these changes on the practice aspects of management.^{6,7} Some changes to long-term condition management may be subtle (e.g. increasing amount of exercise) whereas others may be more evident (e.g. insulin injections), highlighting the need for a longer temporal perspective of management to be understood.⁸

The type and amount of support available for self-management varies over time and may be sought from a range of sources.^{9,10} Yet, long-term condition management does not occur in isolation, rather it is influenced by family, friends, healthcare professionals or work colleagues, i.e. an individual's social network.^{8–12} Social networks are comprised of a complex interplay of network members who provide access to a range of support and information to support long-term condition management.^{9,12–14} This reciprocal influence of social networks on health has remained an underexplored element in considering health as a social process.

Changes to the support available for long-term condition management may be reflected by changes to an individual's personal social network.^{9,11} There is a need to understand what influences changes to everyday long-term management and the influence of social network members within this context. The aim of this paper is to explore how long-term condition management changes over time and the influence of social network members on where and how people seek support.

Method

A longitudinal qualitative social network study was conducted with individuals who had either diabetes, irritable bowel syndrome (IBS), or chronic obstructive pulmonary disease (COPD) in the north-west of England.

Study setting and recruitment

The study was embedded within a formative evaluation which preceded a randomised controlled trial (RCT) which aimed to implement a whole systems approach to self-management support for long-term condition management.¹⁵ Participants were provided information about this study when they attended their General Practice and consented to be contacted. RM contacted potential participants to discuss the study and were told that this study was part of a doctoral study and an embedded qualitative study within a formative evaluation of an RCT. Those who agreed to participate were recruited. The researcher who contacted and conducted the interviews was a female. Interviews were conducted in participants' homes and the interviewer and interviewee were present.

Data Collection

Initial face-to-face interviews, telephone follow-up interviews, and final face-to-face interviews were conducted a year later.

Written consent was obtained from all participants before the initial interview and verbally reconfirmed before the final interview. RM conducted all interviews. All face-to-face interviews were audiotaped and transcribed. Fourteen participants completed telephone follow-up interviews. Field notes were made during telephone calls. The duration of the initial interviews was between 34 to 92 min (mean: 59 min, SD: 16.17), telephone calls from 5 to 16 min (mean: 12 min, SD: 3.11) and final interviews from 44 to 79 min (mean: 62 min, SD: 8.24). Semi-structured interviews explored participants' management strategies, experiences with healthcare providers and the influences of their social network members on their long-term condition management (see Tables 2 to 4).⁹ Participants were asked about the relationships, types of support they received, and what influenced their long-term condition management.

Participants mapped their social networks using diagrams with three concentric circles in the final interviews.^{9,16} Participants placed members in the central circle in response to 'Who do you think are most important in terms of how you manage your health and long-term condition?' In middle circle, participants identified members who were less important than in the middle circle, and members in the outer circle were considered less important than the middle circle. The centre of the social network represented the interview participant (or ego) and the other circles represented their social network members (or alters). The thickness of the lines between the ego and alter on the diagram represented how frequently the participant had contact with the alter (i.e. the thicker the line, the more frequent the contact), and the size of the circle represented the proximity of the alter to the participant (i.e. the larger the circle, the closer the network member lives). The gender of the alters is represented by the inside the circle (e.g. symbols with a cross end represent

female members). Participants were asked to complete the diagram at the start of the interview and given as much time as they wanted to do this (for example this could be a few minutes or the duration of the interview as participants added new network members through the discussion).

Data analysis

Data analysis was ongoing throughout the study. All authors met regularly to discuss emerging themes. All authors analysed transcripts and commented on the interpretation of the data and agreed the key themes and concepts. All names reported were anonymised using relationship codes and pseudonyms. A combined narrative and thematic approach to data analysis was used to allow themes that emerged to be explored across the data set whilst enabling the individual context of management and change over time to be included as a unit of analysis.⁹ Atlas t.i. 5.2 (Atlas.ti, Berlin, Germany) was used to support the labelling and organising of themes. VennMaker version 1.03 (VennMaker, Cologne, Germany) was used to organise and anonymise the network diagrams for analysis. Transcripts were returned to participants for comment and no changes were made to the transcripts by participants. The network diagrams were analysed descriptively to identify who was in the network, the frequency of contact and the distance of alters. Each participant was considered as a case, and their narrative was used to consider the category characteristics, implications and their boundaries. In this study, a narrative was defined as '*an everyday oral storytelling a speaker connects events into a sequence that is consequential for later action and for the meanings that the speaker wants listeners to take away from the story*' (p3).¹⁷ Five case studies will be used as exemplars in the results section to highlight how changes in social networks and changes health management were intertwined to

have either positive, negative or little influence on management. This study adheres to the consolidated criteria for reporting qualitative research (COREQ).¹⁸

Results

Thirty participants were purposefully sampled with an index condition of either diabetes ($n=15$), IBS ($n=8$) or COPD ($n=7$) (see Table 1). Nine people who were approached declined to participate as they did not want to take part in a longitudinal study. Participants were recruited in the north-west of England from two general practices (practice a, $n=19$; practice b, $n=11$). One participant withdrew after five months into the study due to ill health and three were unreachable for final interviews. Fourteen men and sixteen women were interviewed and their ages ranged from 25 to 84 (mean: 59, SD: 15.01). Participants had between one to six long-term conditions (mean number of conditions: 2.7, SD: 1.58).

The types of social networks that participants depicted and co-morbidity emerged as salient issues.^{8,9} The main themes that emerged as salient in the management of long-term conditions over time was the processes or events that occurred when management priorities changed, i.e. critical moments. These factors which influenced the role of social networks which influenced whether there was a positive, negative or neutral change were highlighted in analysis (e.g. expectations of social roles, or availability of support). The relationships that the participant had with their network members were not static as they could change or be redefined. The changing of relationships did not appear to reflect changes in conditions of participants or gender.

Defining critical moments

A critical moment is defined as the development of a critical mass of multiple factors

that either accumulated or alternatively were singular events which acted as turning points, to cause a shift in management priorities amongst other everyday (non-health related) priorities. Examples of critical moments included divorce, declining health, re-establishing broken ties with former network members, and bereavement. To manage the complexity of health management, the participants responded by adopting a process of prioritisation (i.e. the order of importance allocated) which changed over time.

...just watching what I eat now...with diabetes really more than anything like... But this breathing, this COPD that's what will affect me in another few weeks when it starts getting cold

(Jack's first interview, Diabetes, high blood pressure, high cholesterol, male, age 65)

This shift was based on a complex interaction of factors including: timing between diagnoses, prior experiences, recommended self-management activities, bereavement, contact with health services, the relative severity of illnesses and flare-ups of different conditions (see below). The processes of prioritisation and change involved the supplementation (i.e. existing network members adopting the role or activities of another network member) over time as support and resources were sought from the network. For example, Don had recently re-established his relationship with his daughter, Shelly, after years of not being in contact, and described that relationship in terms of it developing. Yet, as this relationship had progressed, Shelly's importance had increased which he showed visually as Don moved Shelly from the outer circle to the middle circle of the network diagram.

Obviously my mum, the people I've named in there besides my step daughter obviously, because she's my wife's daughter who's been part of my family since she was four... Collin and Alan are friends for forty years...

Table 1. Participant demographic information and condition information.

Participant pseudonym	Gender	Age at start of study	Index condition	Co-morbid conditions (self-defined)
Catherine	Female	36	IBS	Occipital neuralgia, reoccurring cystitis
Chris	Male	65	Diabetes	None reported
James	Male	59	Diabetes	High blood pressure, cholesterol
Beatrice	Female	46	Diabetes	None reported
Abbie	Female	53	COPD	IBS, depression
Tina	Female	69	Diabetes	Stress incontinence, eating and sleeping problems, hair loss, eye infections, skin and gum infections
Adrian	Male	82	Diabetes	Rheumatoid arthritis, high blood pressure
Don	Male	48	Diabetes	Cataracts and eye problems, tendonitis
Adam	Male	—	Diabetes	Knee problems, kidney problems
Danielle	Female	66	Diabetes	MS, underactive thyroid, high cholesterol
Natalie	Female	57	IBS	Blood pressure, cholesterol, hypertension, COPD
Lyn	Female	57	COPD	IBS
Leo	Male	51	IBS	None reported
Frank	Male	65	COPD	Hypertension
Tom	Male	52	Diabetes	High cholesterol
Rachel	Female	—	COPD	None reported
Jane	Female	55	Diabetes	Epilepsy
Sarah	Female	31	IBS	None reported
Debbie	Female	62	IBS	None reported
Ron	Male	84	Diabetes	Ischemic heart disease, arthritis
Ted	Male	83	IBS	Hearing problems, high cholesterol, memory problems, back pain, signs of angina (participant wording)
Isabella	Female	50	Diabetes	Chronic depression
Kate	Female	84	COPD	High blood pressure, blackouts
Nancy	Female	76	COPD	Arthritis
Jack	Male	65	Diabetes	High blood pressure, high cholesterol
Todd	Male	44	IBS	None reported
Zac	Male	65	Diabetes	Heart bypass, ulcers on bottom of feet that won't heal
Rita	Female	25	IBS	Anxiety problems
Matthew	Male	69	COPD	Oesophageal problems (caused by a hiatus hernia), feet problems
Donna	Female	54	Diabetes	High blood pressure and high cholesterol

My daughter just, I don't trust her at the moment. She would have been in the outer ring, but slowly she's coming into it...

(Don's third interview, male, Diabetes, Cataracts & Eye problems, Tendonitis, age 48)

Prioritisation was an ongoing process and reflected the changing nature of the significance of conditions. At these transition points, management activities were reordered, support and reassurance were actively sought and influenced whether individuals

Table 2. Baseline interview guide.

1. How would you describe your current state of health? What are the conditions you have? Which one if any has priority at the moment in terms of having to manage it?
2. When did you start to have contact with health services about this?
3. How does having your condition affect your life on a day-to-day basis?
4. How do you manage NOW on a day-to-day basis with your condition?
5. Have you had to make any changes to your lifestyle and your diet?
6. What do you currently do when your symptoms get worse?
7. Starting from when you first thought something was wrong, can you tell me how you have responded to your illness and what sort of adjustments you have had to make to your life and what you do on a daily basis?
8. Are there things that other people tell you should be doing but you don't do? What are these things and how do you feel about other people telling you these things?
9. Have you used any information concerning your condition?
10. Do you speak to the pharmacist at all about medication for long-term conditions?
11. Contact with voluntary organisations concerned with your condition?
12. Who in your family or circle of friends locally do you talk to about your illness? When and what do you talk about?
13. Are there occasions when you prefer not to talk to people and keep things to yourself and if so why?
14. What things in your neighbourhood make it easier to manage having a long-term condition and what things make it difficult?
15. What sorts of contact with people (your family and friends, neighbours, local people) make things easier in managing a long-term condition and what makes things more difficult for you?
16. How long have you lived in the area?
17. Could you describe your neighbourhood to someone who wasn't from the area?
18. Do you feel part of a community?
19. Do you get on with/look out for your neighbours and vice versa?
20. Would you say you know most of the people in your neighbourhood?
21. Does anyone help you? If so in what ways?
22. How many times in the past two weeks have friends or family visited you or you visited them?
23. Who have you talked to about your health?

became overwhelmed and disengaged with condition management. Alternatively, critical moments acted as an impetus for participants to take control over their health. Critical shifts in relationships also acted as turning points whereby existing management was disrupted thereby impacting on everyday practices and management.

Positive change to health management over time

Positive changes in health management identified in the participants' narratives were noted for some participants. The participants described changes to factors which

had inhibited their management and valued social roles such as 'being a grand-parent'. This precipitated a critical moment by prompting action in terms of self-management and utilisation of resources within the network to improve their health management. Alternatively, the reconnection of broken ties with former network members provided access to further support and resources.

The influence of re-evaluating social roles

In accounts that depicted the breaking of close ties through life events, such as divorce and death, there was a sense of distress from

Table 3. Monthly interview guide.

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1. How would you describe your current state of health?
 2. Has there been any changes in your condition in the last month? Have you had any time off work?
 3. Has your condition changed your life on a day-to-day basis in the past month? What has having your condition changed in your life?
 4. Have you made any special changes to your diet or lifestyle in the past month?
 - a. What prompted changes? Have you read anything/seen on TV/Internet?
 - b. Are you able to exercise regularly? What helps/hinders?
 5. What health services have you used in the last few weeks?
 - a. Who have you spoken to? E.g. Pharmacist, NHS direct, smoking cessation programme, condition specific clinic?
 - b. Have you made any appointments?
 6. Who have you asked for help from in your family or friends? If so who, how regularly and what have they been doing? Have you visited any friend or family in the past month or had them visit you? If so who, how often, where did you meet?
 7. Have you spoken to any friends or family about your condition?
 8. Have you been in contact with any voluntary organisations concerned with your conditions? Have you received information or support from any other sources for your condition?
 9. Is there anything important, that we haven't discussed, that has happened in the last month that has affected your health which you would like to mention?
-

these events which led participants to re-evaluate valued social roles, such as 'being a husband.' This re-evaluation in turn influenced engagement and the context in which self-management occurred. Conversely, reconnecting with valued network members through redefining or reframing expectations of the relationship had a positive effect on health. The first case study is of Leo. Leo, who had IBS, in the initial interview described being able to talk to *'my close family like my parents, not anybody else really'*. Over the course of the study, Leo described their relationship becoming increasingly more amicable. In the final interview, he described the importance of re-establishing a relationship with his ex-wife, Dawn, which involved redefining the meaning and expectations that were available from each other. This change and revaluation of roles resulted in him considering his ex-wife as one of the most important social network members (see Figure 1). He would seek her advice about

health concerns before going to the General Practitioner (GP).

...probably Dawn [ex-wife]. I'd probably talk to her about something before I went to ...probably after it [IBS] was diagnosed, because the lead up to it being diagnosed was, sort of, the breakdown of the marriage ... obviously we weren't talking about things...

(Leo's third interview, IBS, male, age 51)

Reciprocal influence of enhancing self-management and social network members

Positive changes to health management were also reinforced by key network members and enhanced engagement with self-management. This shift in engagement of management was preceded by an event which acted as a tipping point (i.e. a point where change occurs) for change. The second case study of Lyn, for example,

Table 4. Final interview guide.

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1. In the inner circle, place those who are most important to you in terms of your health, in the next circle place those who are important but not quite as important as those in the inner circle, and in the outer circle place those who are important but not as important as those in the other circles.
 2. Highlight in the same colour the network members who know each other
 3. How far away do they live/work (write next to name):
 - 1a: co-habiting
 - 1b: short walk/drive away
 - 1c: lives upto one hour away
 - 1d: over one hour away
 4. How much contact do you have with them (write next to name):
 - 2a: daily
 - 2b: at least once a week
 - 2c: at least once a month
 - 2d: every couple of months
 - 2e less often than every couple of months
 5. Who is most important in the network for you? Why?
 6. Compare the different network members and what they mean/their role in management?
 7. Who among the people in your diagram do you help? Why?
 8. Was there anyone who was more important but is less so now?
 9. Is there anyone who was less important but is more important now? Why?
 10. Are there people who are making it difficult in some ways? Some people have said that there are certain people that make it harder for them to manage their condition, is there anyone like that in your diagram?
 11. Who do you socialise with—Are their people who aren't on here who you see regularly? Would you talk to them about your health?
 12. Out of these, who would you talk to about health issues/ask for help? Is this different to the people you spend time with or talk to generally?
 13. Health professionals: Did they mention or not? How do you feel about your relationship? How do you prioritise their role in your health care?
 14. Who do you talk to about your condition? How often? Has this changed over time? How is this different to the people that you are close to or spend a lot of time with generally (do you talk to them about your health? If no-why?
 15. What is your main condition priority at the moment? How has your condition/s changed over the last year?
 16. What things have stayed the same, got better or changed for you over the last x months.
 17. Have you *done* anything differently about your condition over the last x months?
 18. Have you changed the way *you think* about your condition and in what way?
 19. New things you have done: Have you talked to anyone or made contact with any service or local activity over the last year
 20. Have you stopped seeing anyone or doing things you previously did at work home or locally, if so what are the reasons?
 21. More broadly how has the area changed? What is the most important thing in the area that you use and things that you do daily activities—how have these changed? Has this been effected by your health?
 22. Tell me about your contact with primary care over the last few months. For your chronic condition, for other things? Has it changed over the last year? If so how is it different from before? Has this impacted on how you view your condition and support from the service and elsewhere?
-

(continued)

Table 4. Continued

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23. How relevant or important has this change been compared to other changes for you in living with your condition over the last year?
 24. Have you noticed any differences in the priorities of the GP or nurse during the consultations?
 25. How often do you see the GP or nurse in the last 12 months?
 26. In general has anything changed for you in the way in which you use or talk to people in primary care about your chronic condition?
 27. Have you noticed any differences in the way that primary care responds to you or your chronic condition and generally?
 28. If have more than one condition – Did you focus on one condition over the other during your contact with services? Thinking about who you have spoken to about your health, would you talk to them about both of your conditions? Is there any difference in which/what you would talk about?
 29. Are you involved with any voluntary organizations? Have you joined anything locally over the last year? Do you do anything differently in the way in which you talk to people about your illness or what you do on a daily basis including in the work place?
 30. What has been the most significant change for you over the last year?
-

described in the first interview increasingly being restricted from daily activities by her multiple conditions, and especially limited by breathing difficulties from COPD.

Although over the course of the year Lyn's health had deteriorated, indicated by crude markers such as the number of medications she was taking had increased (from 7 to 14 daily), she now framed her health and well-being in positive terms. In the final interview, Lyn described how her health had gone through a period of perceived rapid deterioration; however, she described increasing family support and encouraging her to discuss with her GP her increasing tiredness. Lyn described being so tired and lacking energy that she believed she '*was going to die*' and this critical realisation acted as a tipping point to make her seek help and as a catalyst for action.

A: I've always had big sleep problems...support from the family...yeah this is when I said to him [G.P.] I'm going to get myself motivated...get myself cracking...I kept saying I'll die in the chair if I carry on...

(Lyn's third interview, COPD and IBS, female, age 57)

Negative changes to health management over time

Negative changes to health management over the course of the study were particularly evident in participants' accounts when they occurred around major events which caused a reassessment of roles and expectations of support. The third case is of Jack who identified how his relationship had changed with his brother as a result of him staying with him soon after Jack had had a heart attack. This change in relationship was depicted as a loss in the participant's account. The loss of key network members subsequently influenced the expectations of the remaining network members. This caused a shift in expectations of reciprocal roles and responsibilities from network members which in turn could affect where the participants could seek support.

Familial change and loss influencing health

The fourth case study is of Catherine who initially identified her parents as her most important source of support for information

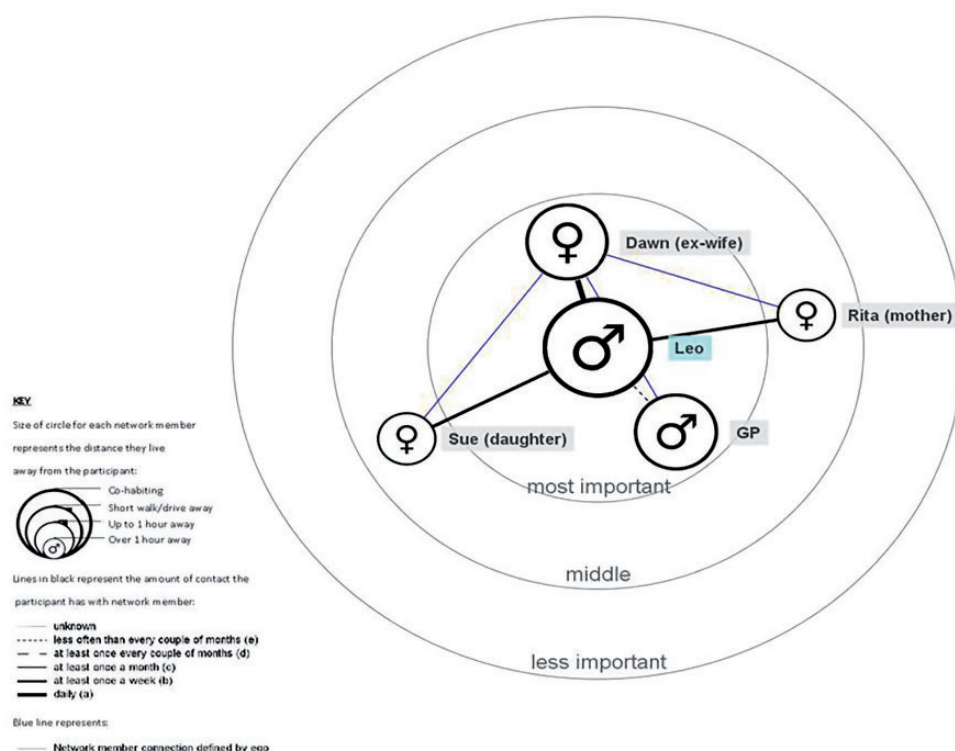


Figure 1. The personal health-related social network of Leo.

and advice. In the first interview, she described being comfortable managing her IBS, which was exacerbated in response to conflict with her ex-husband.

Mum and dad ... They are great ... yeah, my parents are like the most consistent people in my life really, so I speak to them, and they can always tell, cos they've seen it for thirty six years, they can always tell when I've not been to the loo ...

(Catherine's first interview, female, IBS, occipital neuralgia, reoccurring cystitis, age 36)

However, over the course of the study, first Catherine's brother died and subsequently her father died a few months later. Increasingly during the telephone follow-up interviews, she described a

series of illnesses which she linked to having not looked after herself through these periods of bereavement. The impact of these losses was relative, as Catherine had a close relationship with her father and would call on him for support. Conversely, the loss of her brother, John, whilst traumatic, was more expected because he had had a series of serious health problems. This had less impact on her health management as she had not sought support from him.

... My dad should be next to me mum really because I could speak to him about anything ... I wouldn't say my brother because we were never ... since John [brother] died and since my dad died I've been ill about four times, not just irritable bowel.

(Catherine's third interview, female, IBS, occipital neuralgia, reoccurring cystitis, age 36)

Changing roles of remaining social network members as well as the participants' roles in relationship also impacted on the availability of support, following the loss of key network members. With the loss of her father, Catherine described increasingly taking on responsibility, such as shopping, for her mother which led to perceived tensions developing in their relationship.

I would talk to me mum . . . I do feel that I need to go and make sure she's okay, because I would never forgive myself if I didn't and something happened . . . I do take a lot of her stresses on . . . I do get very frustrated by her because she's kind of, given up . . . if I'm not there to take her anywhere she doesn't go and she'll go without bread or milk . . .

(Catherine's third interview, female, IBS, occipital neuralgia, reoccurring cystitis, age 36)

The loss of her father and brother made Catherine re-evaluate the support that she had available through her network. She described '*I don't have friends that I just phone*'. Instead she identified that her friendships were not based on a reciprocal level of support through her friends' actions around the loss of her brother and father.

It's sad because my cousin, who is in the outer circle is a nurse and she's lovely but she is very wrapped up in her own life, whereas I think I'm here for anyone . . . like no one comes here, no one visits . . . if I didn't have Sam [partner] I would have no one . . . it's made me concentrate a lot on me now because I was always there for them . . . I've got to admit that maybe the likes of Vera and Mary have got me so riled in the past that it has made me ill . . .

(Catherine's third interview, female, IBS, occipital neuralgia, reoccurring cystitis, age 36)

Health management remained the same over time

Alternatively some participants described their health management and social networks in relatively stable terms. For some participants, this meant there was little or no reported change to their health or social network over the year. This did not mean that there were no health problems or changes in their health, but rather it did not cause disruption to their ongoing management as changes were concordant with existing practices. Furthermore, this stability was reflected in the way in which the key social network members maintained and consolidated links, despite changing health circumstances. These participants had learnt to manage their condition and normalise practices into their everyday lives and during the study the way they framed their condition had not been threatened or existing notions of management challenged.

. . . I know that if you spoke to them they would help . . . I'm sure they would, but I get on with it and I'm a practical person, I just, this is what I've got and you get on with it . . .

(Chris's third interview, Diabetes, male, age 65)

Normalised health management over time

The fifth case study is of Donna who is an example of how changes in health, through a knee problem which stopped her from working for six months, led to a reconstruction of her health narrative. In the initial interview, Donna described how she had normalised her management into her everyday routines through support mainly from her partner, close friends, and people from work. She was confident managing the diabetes through diet and exercise.

I'm not one of these that takes to me bed . . . I just carry on with my day-to-day life it's

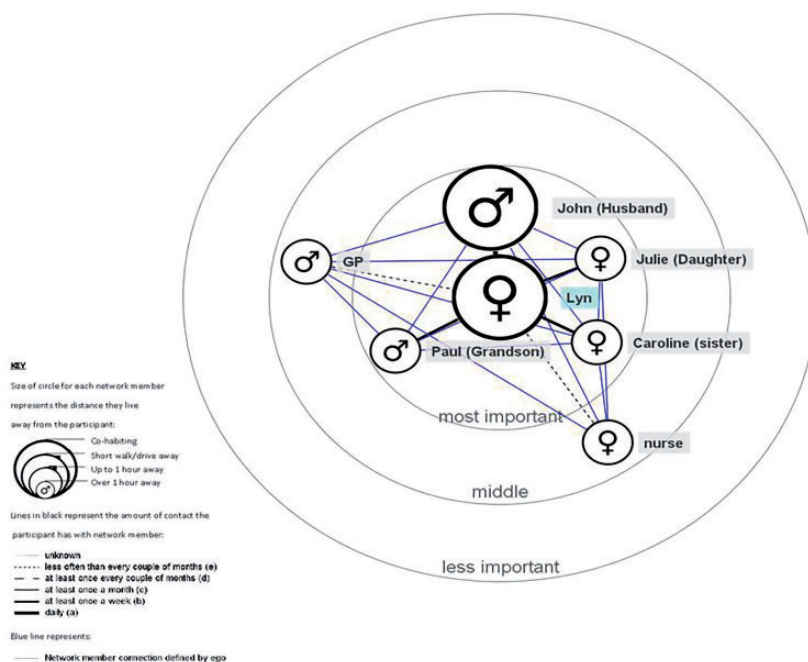


Figure 2. The personal health-related social network of Donna.

not changed any... I've not made a big deal of it at all... after the initial diagnosis, I just got on with it; did what I had to do...

(Donna's first interview, diabetes, high blood pressure, high cholesterol, female, age 54)

In the final interview, Donna's description of sources of support and how she framed health management remained similar to the initial interview (see Figure 2). Her main source of support remained her partner, work colleagues, and close friends despite having to have knee surgery and being bed-bound for six months.

I tend to cope with myself. I don't talk about my health really. It's Peter mainly, and Doris... and the girls at work... so I'm not going to talk about my illnesses socially... It doesn't embarrass me in any way... If I needed to speak to somebody about it, I would do... but on day-to-day support, it's

your friends and family... It's not a major deal; it's not a big part of my life...

(Donna's third interview, diabetes, high blood pressure, high cholesterol, female, age 54)

Despite the change in accessing support from Donna's work colleagues, who she described as her 'extended family', she was able to manage the emotional effect of the injury through the adaptability of the network members and the reciprocal value of their relationships which influenced her motivation to do self-management activities (e.g. lose weight).

Discussion

Summary of main findings

The context in which health management occurs involves embedding management practices into everyday settings through ongoing processes of negotiation and

prioritisation. Throughout the study, the changes in participants' narratives depicted an ongoing reassessment in their understanding of, and engagement with, self-management practices which was reciprocally influenced by their social network.

Support was sought through processes of substitution and supplementation across the social network depicting a re-evaluation of available support. Conversely, for some participants, despite changes in their health or social networks, the meaning of management remained relatively unchanged. Changes in the social networks over time could be considered to exist on a continuum between positive and negative reciprocal influences with health management. There was an ongoing process of breaking or reconnection of ties within the social networks which shaped the context of health management and the degree of individual engagement with management practices.

Comparison with existing literature

Critical moments in participants' accounts occurred when changes in relationships acted as turning points. Turning points occurred when existing management was disrupted which impacted on everyday self-management. This was similar to Denzin's^{19,20} description of epiphanies, which defined an epiphany as '*interactional moments and experiences which leaves marks on people's lives*'.²⁰ In this study, the phrase 'critical moments' was used to similarly reflect the accumulation of a critical mass of seemingly small changes to the context of management, or single large events which acted as turning points, to cause a shift in management priorities. It is through these turning points that change occurred.²¹ Changes to social networks prompted the reassessment of trust and meaning of existing relationships and could act as a catalyst for changes through seeking information and resources from different network

members. Relationships cyclically influenced the meaning of management which shaped the context in which management occurred. Relationships which had negative as well as positive influences remained included in the social network because of the value of certain types of support provided and additional support needs were substituted by other network members. However, when the flexibility in networks was limited, for instance because of the death of a key figure, this challenged existing self-management priorities. Complex relationships, for instance with partners or close friends, could influence individual management on multiple levels.

The importance of relationships was characteristically defined by reciprocity of support and levels of trust developed over time. Accounts depicted how the expectations of reciprocal support shaped what type of support would be expected. Similarly, Pescosolido^{11,22} proposed a Network Episode model which had two core characteristics: first, that the illness career was dynamic and second that social networks had a role in shaping health care outcomes. There are parallels with this model with the ongoing process of decision-making in long-term condition management. By examining the changing influence of social networks on long-term condition management over time, we have illuminated the influence of changing support needs in response to long-term condition management.

Change in network membership may have wider ranging effects than simply the loss, or reestablishment, of relationships as it prompted the reassessment of trust and meaning of existing relationships. Support was sought through processes of substitution and supplementation across the network and was highlighted around critical moments that lead to participants depicting a re-evaluation of available support. The help available from network members varied

over time and support was sought from a range of ties. Certain members of the network were more important at certain times. Changes were found to influence how participants managed their condition and whether or not they sought other sources of support.

There is a growing recognition of the importance of social networks on long-term condition management.^{9,12,13} This study has gone further than previous studies to identify how the social networks implicated in *ongoing* long-term condition management changed overtime. This builds on the literature which examines the role of life-events in the initial genesis and diagnosis of long-term conditions. Participants' accounts across a range of ages showed processes of breaking, reconnecting and establishing new ties with network members. Previous studies have shown that in response to decreasing health, the size of the network remains approximately the same but the amount of support provided by the network members increases.^{23,24} The findings here suggest that it is more nuanced. Rather than the size of the social network being important, it is the type of support that is offered based on the reciprocal and ongoing relationship with network members.

Implications for future work and clinical practice

The findings of this study have further implications in considering the relevance of measuring social networks quantitatively. Relationships were complex, often with both positive and negative components, and the size of the network did not necessarily relate to its ability to provide support. This also has implications in the wider social network literature as it challenges the presumption that as individuals age, the size of the social networks decreased, for instance through retirement or illness. Other sources of less formal support (e.g. voluntary and community groups) may provide alternative

support and the flexibility to provide different support at different times in response to changes in health management. Further work could explore the role of referral from primary care to local community groups. Furthermore, whilst there is an increasing recognition of the role of carers and informal sources of support, this leads to questions about what happens when someone is not able to substitute for support and there remains a need to understand the role of loneliness on self-management.

Strengths and limitations of the study

This study had a longitudinal design, which enabled themes that arose to be explored further with the same participants. By using research methods which enabled individual's narratives to guide interviews, salient topics arose and a more complex and complete view of condition management, individual priorities and the strategies employed to embed it within the everyday was obtained. This enabled the identification of the relationship between health and social networks evolved overtime to be considered and the relative importance and role of various actual and potential sources of support. This is important when considering the appropriateness and provision of formal self-management support programs. The nature of longitudinal research allows for emergent topics to be explored and a degree of flexibility in data gathering.²⁵ One limitation of this study is that the social network diagrams were only used for data collection in the final interview as it became apparent from the initial analysis that central network members were being discussed. The diagrams were used in the final interview to draw out the nuances in relationships that were difficult to elicit in the initial discussion because of their relatively more minor role. The ensuing interview discussion helped to develop an understanding of the relative importance and role of various actual and

potential sources of support, which highlighted the network process of substitution and supplementation. Another limitation of the study was that it was not feasible for data saturation to be reached within this longitudinal qualitative study as new themes emerged over time and it was not possible to recruit further participants to explore these and reach data saturation. Another limitation of the study was that 26 of the original 30 participants remained in the study and completed the data collection. Whilst attrition is a common problem in longitudinal studies,²⁵ the initial number of participants recruited was based on the presumption that there would be a drop-out over the course of a year. It can be argued that 'all research is interpretative'.²⁶ As all analysis is based on interpretations which are shaped not only by the context in which it occurs but also by the researchers' own world views, there is the potential for the data to be interpreted differently by someone else. To increase trustworthiness in the data collection, detailed accounts have been given about the data collection analysis and interpretation process. RM collected data and interacted with participants and as the attrition rate of the participants in the final interview was only four, this may suggest that RM established a rapport with participants to support their participation.

Conclusion

By taking a more socialised perspective that considers how social networks influence orientation to management, this study moves beyond existing research by examining the specific processes of support and resource provision, highlighting its complex and reciprocal interaction on self-management. Formal support needs to be tailored to acknowledge the different types of support that may be more varied and wider ranging than clinical services.

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